

Exact K2P and K3P Tree–Theta–Trinet Collisions

Confidential technical summary – not yet publicly circulated

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Exact finding under confidential review. A binary semi-directed strict level-two theta trinet with a genuine nontrivial 3-blob shares an exact K2P distribution with a three-star tree. Under the conventions stated in arXiv:2607.12919v2, this appears to conflict with the conclusion of Lemma 5.6 and with the K2P case of Corollary 5.8. The present construction does not test the paper’s JC or level-one results. A companion exact K3P collision answers its open high-level K3P trinet question negatively.

Common topology. Use

$$\rho \rightarrow 1, \rho \rightarrow u, u \rightarrow p, q, p, q \rightarrow r_2, p, q \rightarrow r_3, r_2 \rightarrow 2, r_3 \rightarrow 3,$$

with both inheritance probabilities $1/2$. Root suppression gives paths $p - u - q$, $p - r_2 - q$, $p - r_3 - q$ and three incident leaf edges.

Simple exact K2P witness. In order (A, C, G, T) , assign

$$\begin{aligned} K &= (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), & U &= (1, \tfrac{4}{5}, \tfrac{19}{30}, \tfrac{4}{5}), & V &= (1, \tfrac{7}{240}, \tfrac{239}{360}, \tfrac{7}{240}), \\ S &= (1, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}), & T &= (1, \tfrac{1}{3}, \tfrac{1}{27}, \tfrac{1}{3}). \end{aligned}$$

Use K on the four pendant rooted edges, U, V on $u \rightarrow p, u \rightarrow q$, and S, T on arcs from p, q . Every transition entry is positive; the smallest is $1/120$. For $x = y + z$ let

$$M_{y,z} = \tfrac{1}{4}(S_y S_z U_x + S_y T_z U_y V_z + T_y S_z U_z V_y + T_y T_z V_x).$$

With $s = \sqrt{71}$,

$$P = (1, \tfrac{151}{36s}, \tfrac{107}{162}, \tfrac{151}{36s}), \quad R = (1, \tfrac{s}{40}, \tfrac{31}{120}, \tfrac{s}{40}),$$

and exact arithmetic gives $M_{y,z} = P_{y+z} R_y R_z$. The tree vectors are

$$\alpha = (1, \tfrac{151s}{10224}, \tfrac{107}{648}, \tfrac{151s}{10224}), \quad \beta = \gamma = (1, \tfrac{s}{80}, \tfrac{31}{240}, \tfrac{s}{80}).$$

All 64 Fourier and pattern probabilities agree exactly; the minimum is $1188799/79626240 = 0.01492973924\dots$. The source invariant Q is zero.

Proof diagnosis. The level-one sign factorization is valid after placing the active reticulation leaf in a designated position. In the level-two induction a child can retain a reticulation adjacent to another leaf. Positivity after a child-specific relabelling does not imply positivity of the fixed-order child invariant in the parent expansion. On the exact continuous-time witness, one child has $Q = -1.919971072382827\dots \times 10^{-9}$ in the parent order and $Q = 3.428488326525925\dots \times 10^{-9}$ after the favorable swap. A separate fully specified rational theta point has $Q < 0$ in all six orders.

K2P geometry and continuous time. The affine K2P space has dimension 9 and the tree model dimension 6. At the simple collision, the fixed theta map has determinant

$$-\frac{7^2 11^2 19 107 151^2 15013}{2^{60} 3^{25} 5^{10}} \neq 0.$$

It also has rank 9 at the strict continuous-time collision. Hence the local semi-directed theta collision locus has dimension 17 and codimension 3: generic theta parameters are distinguishable, but full distinguishability fails. An exact symmetric ansatz contains a local six-dimensional collision family. The strict continuous-time witness satisfies $g > s^2$ on every edge, has minimum eigenvalue $1/16$ and minimum rate margin $11/900$, and is verified by direct Markov pruning.

K3P companion. Let $h = 5^{-1/4}$. Use

$$U = (1, h/3, h, 1/3), \quad V = (1, h, h/3, 1/3), \quad S = (1, 3h^2/4, 1/4, 3/10), \quad T = (1, 1/4, 3h^2/4, 3/10).$$

Its core factors with

$$B = (1, h^2/2, h^2/2, h^2/2), \quad P = (1, (5h^3 + h)/4, (5h^3 + h)/4, h^2).$$

The fixed theta map has rank 15 with

$$\det J_* = \frac{h(10h^2 + 1)}{2^{61} 3^4 5^{14}} > 0.$$

Its local collision locus has dimension 23 and codimension 6. A real-analytic implicit-function argument, with Jacobian and tangent data verified exactly, moves the collision into the cone where all three K3P rate classes are positive.

Status and replay. No public posting or submission has occurred. Run `python3 verify_k2p_simple.py` for the compact witness or `python3 verify.py` for every check; the latter ends with `ALL EXACT CHECKS PASSED`.